

AI Usage Log Entry - Group 16 (Final Submission)

Entry Metadata

- Date: 2026-05-01
- Contributor: Group 16
- Milestone: Final Submission (Week 15)
- Tool(s): Codex (GPT-5 coding assistant)

Prompt Summary

- Objective:
 - Start final milestone implementation work.
 - Add final demonstration automation and artifacts.
 - Prepare final validation, report, and presentation support files.
- Key prompt(s):
 - "Can we now start work on the final milestone?"

Generated Output Summary

- Added final demo runner:
 - `src/validation/run\_final\_submission\_demo.py`
 - `scripts/run\_final\_demo.sh`
- Generated final validation artifacts under `docs/validation/`:
 - `final\_ieee14\_powerflow\_summary.csv`
 - `final\_ieee14\_fault\_summary.csv`
 - `final\_ieee14\_fault\_currents.csv`
 - `final\_ieee14\_post\_fault\_voltages.csv`
 - `final\_submission\_summary.txt`
- Added final validation report:
 - `docs/validation/Final\_submission\_validation\_report.md`
- Added presentation/demo artifacts:
 - `docs/presentation/final\_demo\_script.md`
 - `docs/presentation/final\_presentation\_outline.md`
- Added final report draft:
 - `docs/reports/group16\_final\_report\_draft.md`

Verification and Modification

- How results were checked:
 - Ran `./scripts/run\_final\_demo.sh --fault-bus 4 --tolerance 1e-6 --max-iterations 200`.
 - Verified PASS summary and generated CSV artifacts.
 - Ran full tests (`pytest`) to confirm no regressions.
- What was changed by the team:
 - Kept final demo focused on required IEEE-14 power flow + 3PH fault evidence.
 - Separated final demo artifacts from M1-M5 artifacts for clarity.
- Why changes were needed:
 - Final week grading requires clean, reproducible demonstration evidence and clear submission packaging.

Final Outcome

- Files affected:
 - `src/validation/run\_final\_submission\_demo.py`
 - `scripts/run\_final\_demo.sh`
 - `docs/validation/Final\_submission\_validation\_report.md`
 - `docs/validation/final\_ieee14\_powerflow\_summary.csv`
 - `docs/validation/final\_ieee14\_fault\_summary.csv`
 - `docs/validation/final\_ieee14\_fault\_currents.csv`
 - `docs/validation/final\_ieee14\_post\_fault\_voltages.csv`
 - `docs/validation/final\_submission\_summary.txt`

- `docs/presentation/final\_demo\_script.md`
- `docs/presentation/final\_presentation\_outline.md`
- `docs/reports/group16\_final\_report\_draft.md`
- Impact on project:
 - Establishes a reproducible final-demo workflow and submission-ready evidence baseline for Week 15.